

PORTFOLIO CASE STUDY • SQL + POWER BI

Social Media Engagement Analysis

An end-to-end analytics case study — from SQL querying to dashboard design — evaluating engagement performance across four platforms and four content formats.

DATASET

200 Social Posts

TOOLS USED

SQL • Power BI

PREPARED

09 September 2026

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Headline Metrics (Full Dataset)

200 TOTAL POSTS	4 PLATFORMS	4 CONTENT TYPES	5.05M TOTAL CLICKS
1.01M TOTAL LIKES	195K TOTAL COMMENTS	155K TOTAL SHARES	50.5 / 49.5 SENTIMENT SPLIT (%)

Executive Summary

Purpose. This case study analyses social media engagement performance across four platforms — Facebook, Instagram, YouTube, and LinkedIn — and four content formats, using SQL for data querying and Power BI for dashboard visualisation. It is structured to demonstrate a complete analytics workflow: from raw schema through query design to business-ready findings, on a dataset of 200 posts.

Data source and approach. The underlying dataset is a single-table social media engagement log (schema detailed in Section 2) queried with a 25-question SQL script covering aggregation, ranking, and comparative logic. Every figure in this report is drawn from the resulting Power BI dashboard, with each finding traceable to the specific SQL query that produced it.

Top three findings: First, Instagram is the standout platform overall — it leads in total follower growth (104,140) and drives the largest share of clicks of any platform (33.22%, 1.68M of 5.05M total), while YouTube posts the highest engagement rate (8.97%) despite having the lowest average daily reach (30,436) of the four platforms — a reach-versus-engagement trade-off worth understanding before reallocating budget. Second, at the content level, Instagram Article is the single best-performing content type in the visible data, leading on likes (105,808), comments (16,594), clicks (572,029), and shares (17,079) simultaneously — a rare case of one format dominating every engagement metric at once. Third, sentiment is almost perfectly split (50.5% negative vs. 49.5% positive across 200 posts) — engagement volume is strong, but it is not accompanied by a clearly positive audience reception, which tempers how the volume metrics should be read.

Taken together, the data supports doubling down on Instagram and its article format specifically, while treating YouTube's high engagement rate as a signal worth testing at greater reach. The near-even sentiment split is the one finding that cuts against an otherwise positive volume story, and is flagged for follow-up in Section 6.

Dataset & Project Overview

The dataset is a single table, `SocialMediaEngagement`, with 200 rows (posts) and 13 columns spanning platform identity, reach and funnel metrics, content classification, raw engagement counts, and a sentiment label. The schema, as defined in the accompanying SQL script, is:

COLUMN	TYPE	ROLE IN ANALYSIS
Platform	VARCHAR(50)	Grouping dimension — 4 levels: Facebook, Instagram, YouTube, LinkedIn
ContentType	VARCHAR(50)	Grouping dimension — 4 levels: Post, Story, Reel, Article
EngagementRate	DECIMAL(10,2)	Primary efficiency metric
FollowersGrowth, AvgDailyReach, Impressions, Clicks, ProfileVisits, Followers	INT	Funnel and reach metrics
Likes, Comments, Shares	INT	Raw engagement counts
Sentiment	VARCHAR(20)	Positive / Negative classification

Coverage

The dashboard's sentiment visual confirms a total of **200 posts** in the dataset, split 50.5% Negative to 49.5% Positive. Across all platform and content-type breakdowns used in this report, no missing or null groupings were observed in the visualised output.

Data-visibility note. The dashboard's platform-by-content-type matrix is a scrollable table; this static export captures the top 11 of 16 possible Platform × ContentType combinations (ranked by Sum of Likes). Five combinations — Instagram-Story, Facebook-Article, YouTube-Reel, LinkedIn-Post, and LinkedIn-Article — are not visible in the export and are therefore excluded from the content-type comparisons in Track C, though they are included in the dataset-wide totals (Likes, Comments, Clicks, Shares) shown separately, which are computed independently of the visible rows.

- Internal consistency check:** the "Clicks" stage value in the Engagement Value by Stage chart (5,047K) matches the dataset-wide Total Clicks figure (5,047,472) to the nearest thousand, and the four platform shares in the Sum of Clicks donut (1,677K + 1,300K + 1,206K + 865K ≈ 5,048K) reconcile with both — confirming the platform-level and dataset-wide click totals are drawn from the same underlying aggregation.
- Followers Growth figures** shown on the platform KPI cards are presented exactly as labelled on the source dashboard; the dashboard does not indicate whether this is a sum or average per platform, so this report treats it as given rather than assuming one or the other.

Analytical Approach: SQL Methodology

All 25 questions behind this dashboard were answered directly in SQL before being visualised in Power BI. The query set progresses from simple aggregation to comparative and ranking logic, demonstrating a layered analytical approach:

1 DESCRIPTIVE Counts, averages, and totals by platform and content type (Q1-Q11).	2 COMPARATIVE Rate metrics — CTR, profile-visit rate — computed as ratios of sums, not averages of ratios (Q15-Q18).	3 CONDITIONAL Correlated subqueries via HAVING to flag platforms that beat one metric but lag another (Q19-Q20).	4 RANKING Window functions to find the best content type per platform, and top-N within each group (Q22-Q23, Q25).	5 VISUALISATION Every query result mapped to a specific Power BI visual for the final dashboard.
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Two techniques are worth highlighting as they go beyond basic aggregation:

CONDITIONAL FILTERING WITH A CORRELATED SUBQUERY (Q19)

```
-- Platforms with high reach but below-average engagement
SELECT Platform,
       ROUND(AVG(AvgDailyReach), 2) AS Avg_Reach,
       ROUND(AVG(EngagementRate), 2) AS Avg_Engagement_Rate
FROM   SocialMediaEngagement
GROUP BY Platform
HAVING AVG(AvgDailyReach) > (SELECT AVG(AvgDailyReach) FROM SocialMediaEngagement)
      AND AVG(EngagementRate) < (SELECT AVG(EngagementRate) FROM SocialMediaEngagement)
ORDER BY Avg_Reach DESC;
```

This pattern — a HAVING clause referencing two independent correlated subqueries — flags any platform that is simultaneously above-average on one metric and below-average on another, without hard-coding either threshold. The same pattern is reused in Q20 to find high-engagement, low-growth platforms.

BEST-IN-GROUP RANKING WITH A WINDOW FUNCTION (Q22)

```
-- Best-performing content type for each platform
WITH ContentPerformance AS (
  SELECT Platform, ContentType,
         AVG(EngagementRate) AS Avg_Engagement_Rate,
         ROW_NUMBER() OVER (
           PARTITION BY Platform
           ORDER BY AVG(EngagementRate) DESC
         ) AS rn
  FROM   SocialMediaEngagement
  GROUP BY Platform, ContentType
)
SELECT Platform, ContentType, ROUND(Avg_Engagement_Rate, 2) AS Avg_Engagement_Rate
FROM   ContentPerformance
WHERE  rn = 1
ORDER BY Platform;
```

Partitioning ROW_NUMBER() by Platform finds the top content type *within each platform independently* in a single pass, rather than requiring four separate queries. Q23 extends this to a top-3 using DENSE_RANK(), and Q25 applies the same pattern to find each platform's highest-click content type.

Detailed Findings by Analytical Track

Track A — Platform-Level Performance

No single platform leads on every metric — each of the four shows a different strength, which is itself a finding.

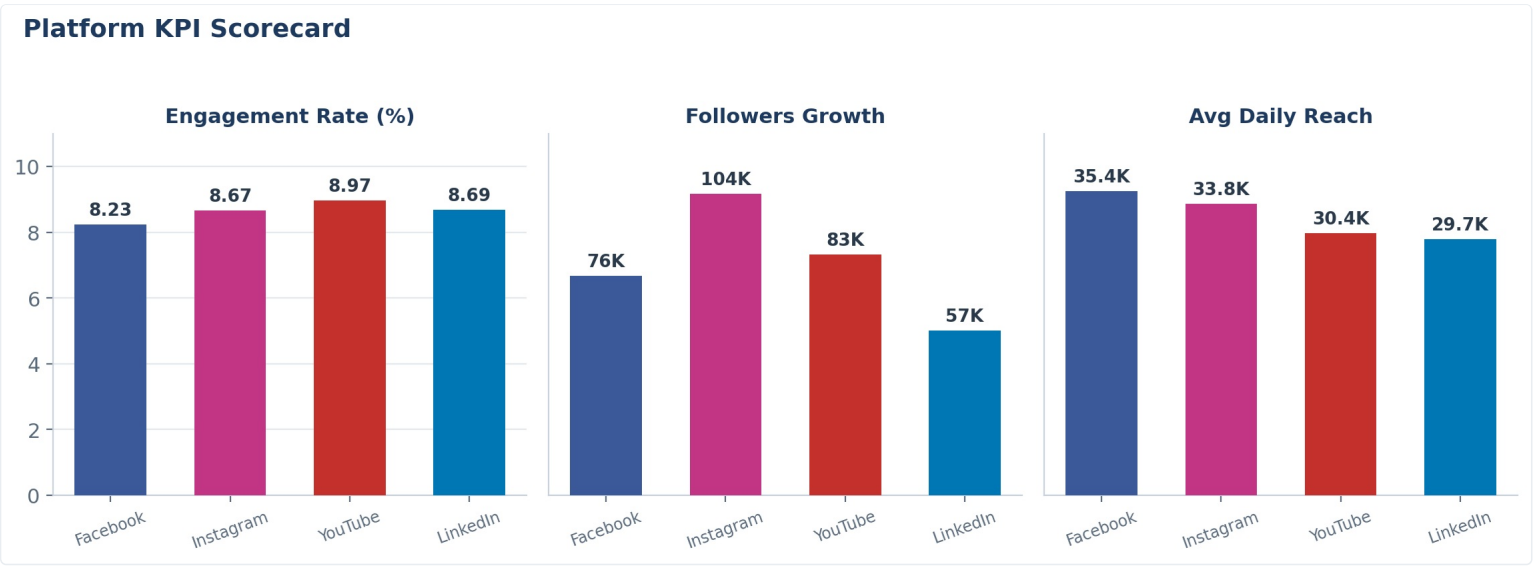


Figure 1. YouTube posts the highest engagement rate (8.97%) but the lowest average daily reach (30,436); Instagram leads decisively on followers growth (104,140), roughly 37% ahead of the next-highest platform (YouTube, 83,271).

Platform	Engagement Rate	Followers Growth	Avg Daily Reach
Facebook	8.23%	75,920	35,350.52
Instagram	8.67%	104,140	33,847.82
YouTube	8.97%	83,271	30,436.12
LinkedIn	8.69%	57,000	29,733.32

What this means in simple terms: Facebook has the biggest audience reach but the lowest engagement rate of the four — it's reaching people without exciting them as much. YouTube is the opposite: smaller reach, but the content that does get seen performs best per-impression.

Track B — Engagement Funnel: Value by Stage

The four funnel-adjacent stages tracked on the dashboard do not decrease monotonically — Impressions and Followers are both far larger than Clicks and Profile Visits, reflecting that these are cumulative or platform-level counts rather than a strict sequential funnel of the same user cohort.

Engagement Value by Stage

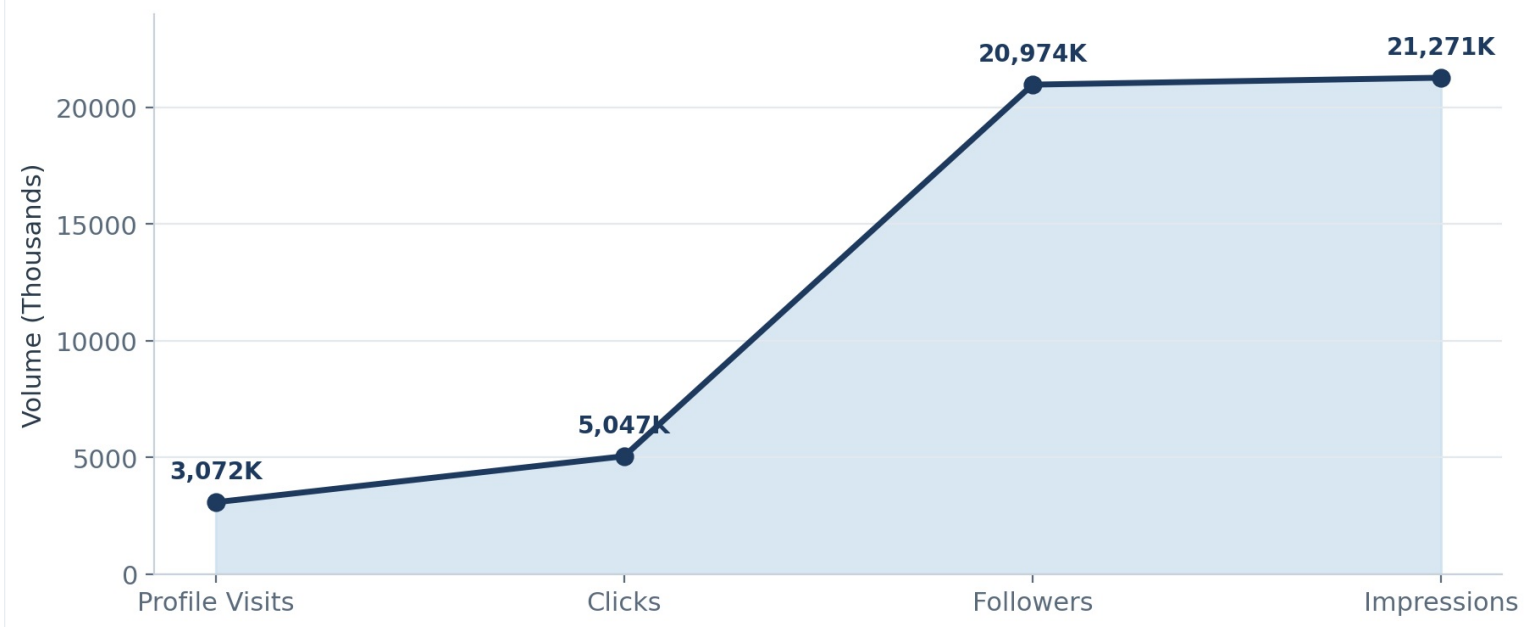


Figure 2. Profile Visits (3,072K) and Clicks (5,047K) are an order of magnitude smaller than Followers (20,974K) and Impressions (21,271K) — the latter two accumulate across the full follower base and posting history rather than being driven by individual post clicks.

What this means in simple terms: don't read this chart as "only 3,072K of 21,271K impressions converted to a profile visit" — Followers and Impressions are large, slow-moving totals, while Clicks and Profile Visits are the more immediate, action-oriented metrics worth optimising day to day.

Track C — Content-Type Performance

Among the 11 visible Platform × ContentType combinations (see the data-visibility note in Section 2), Instagram Article is the standout: it ranks first on every one of the four engagement metrics shown.

Sum of Likes by Platform - Content Type

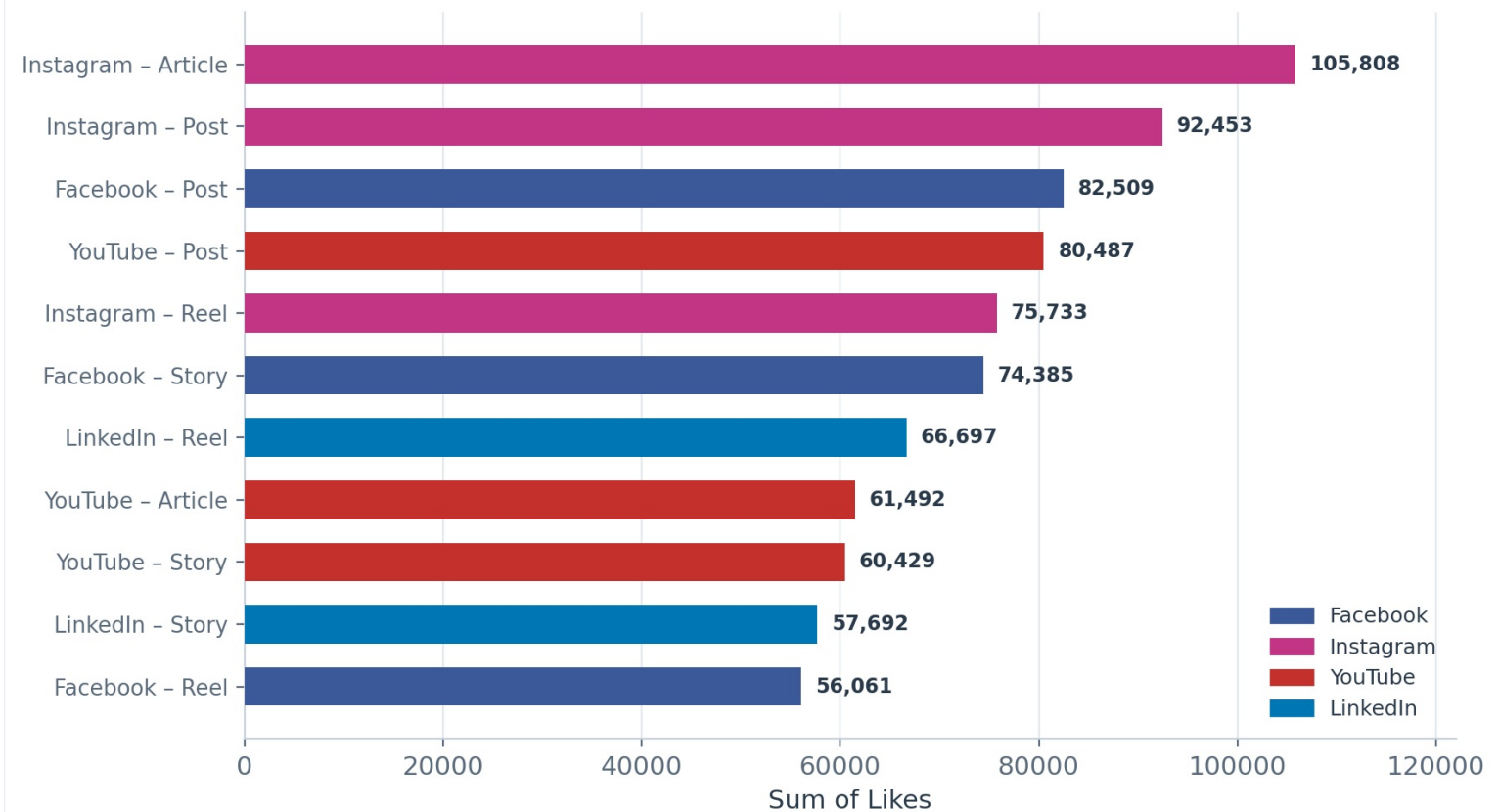


Figure 3. Instagram Article leads all visible combinations on Likes (105,808), roughly 14% ahead of the next-highest (Instagram Post, 92,453).

Sum of Clicks by Platform - Content Type

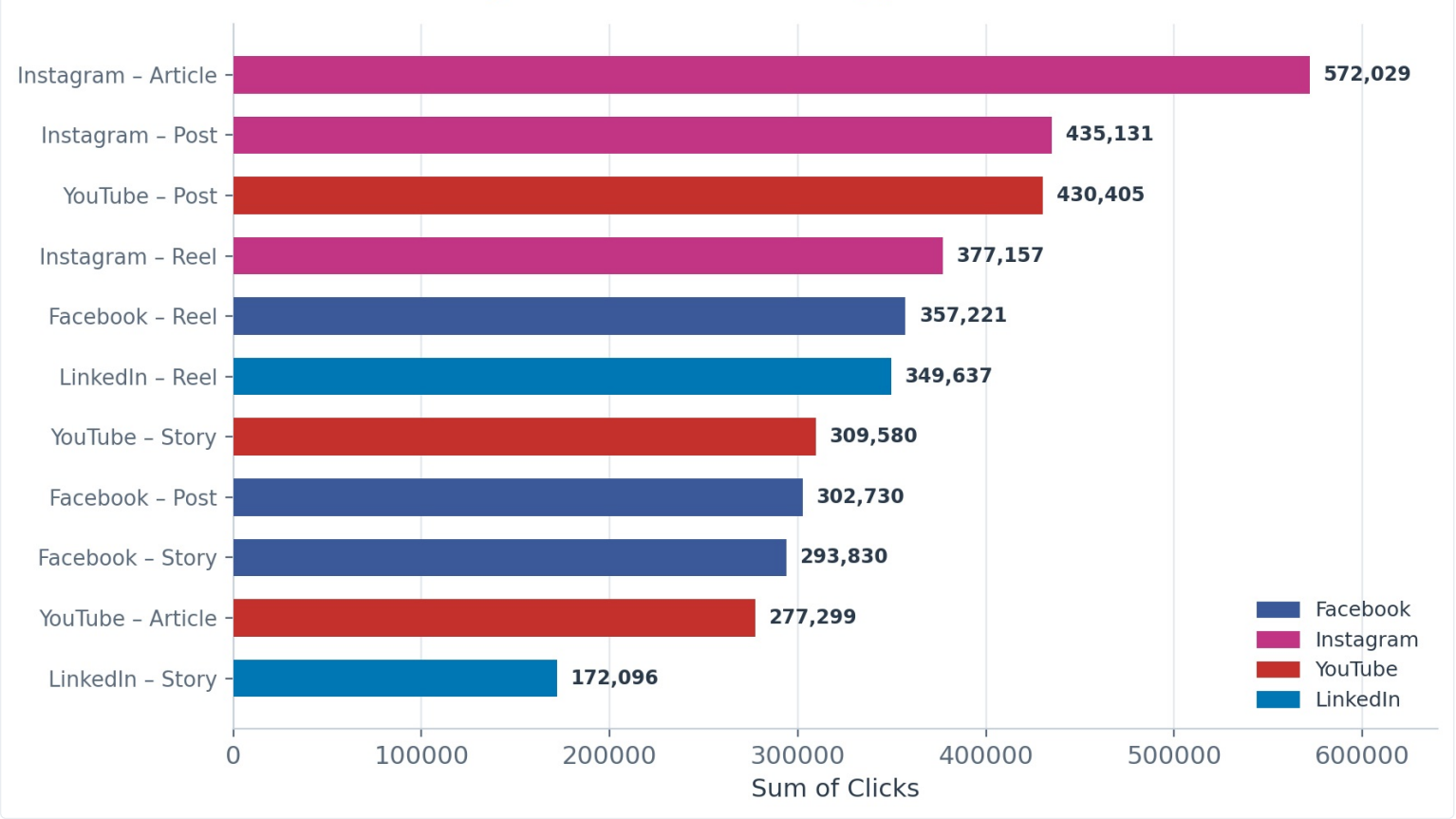


Figure 4. The same Instagram Article row also leads on Clicks (572,029) — the only content type in the visible data to top both a volume metric (Likes) and an action metric (Clicks).

PLATFORM - CONTENT TYPE	LIKES	COMMENTS	CLICKS	SHARES
Instagram - Article	105,808	16,594	572,029	17,079
Instagram - Post	92,453	10,632	435,131	13,719
Facebook - Post	82,509	15,306	302,730	6,616
YouTube - Post	80,487	12,132	430,405	13,676

What this means in simple terms: long-form, article-style Instagram content isn't just popular — it's the format most likely to drive an actual click-through, not just a passive like. That combination (high likes *and* high clicks) is rarer than it sounds, since the two don't always move together.

Track D — Click Distribution by Platform

Instagram's lead extends to click share: it generates roughly one in every three clicks across the dataset.

Sum of Clicks by Platform

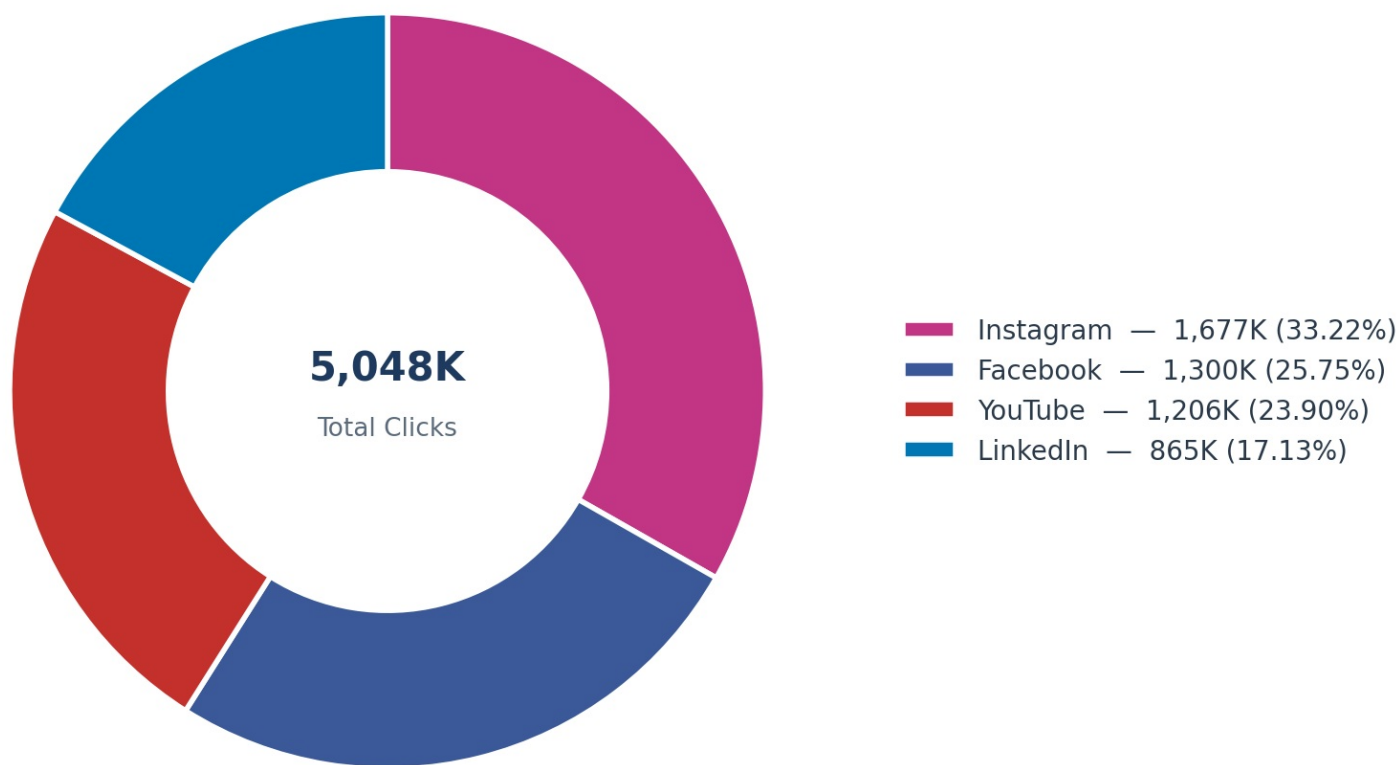


Figure 5. Instagram (33.22%) and Facebook (25.75%) together account for almost 59% of all clicks; LinkedIn trails at 17.13%, consistent with its lower reach and follower-growth figures in Track A.

What this means in simple terms: click activity is concentrated on two platforms — Instagram and Facebook — while LinkedIn, despite being a distinct professional audience, contributes the smallest share of click-through activity of the four.

Track E — Sentiment Overview

Audience sentiment across the full 200-post dataset is essentially a coin flip.

Sentiment Distribution

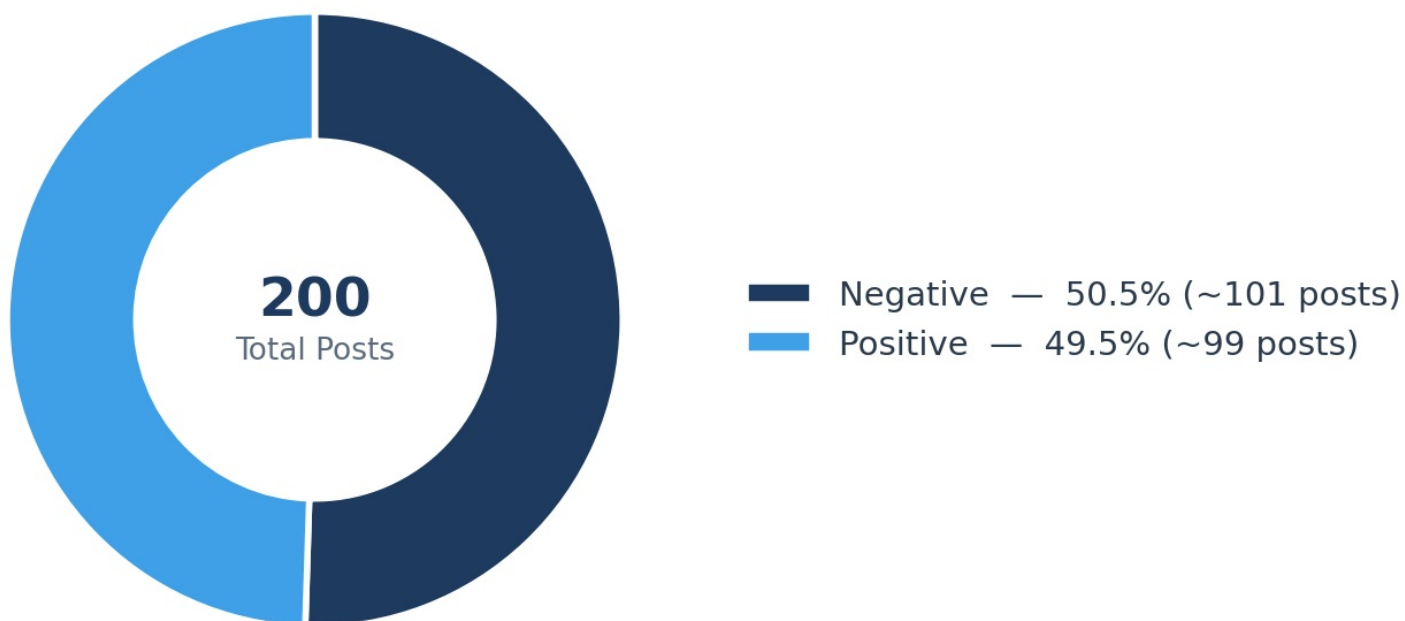


Figure 6. 50.5% of posts (~101) carry negative sentiment against 49.5% (~99) positive — a 1-point gap that is not a meaningful skew in either direction.

What this means in simple terms: high engagement volume does not necessarily mean audiences like what they're seeing — with sentiment split almost exactly down the middle, roughly half of all engaged posts are associated with negative sentiment, which is a separate problem from reach or click volume.

Cross-Domain Insights and Implications

Viewed together, the platform, content, and sentiment tracks reveal patterns that are not obvious from any single analysis in isolation.

Pattern 1 — Instagram's Lead Is Consistent Across Every Track

Instagram appears first or second in every track in this report: highest follower growth (Track A), largest click share (Track D), and home to the single best-performing content type, Instagram Article (Track C). This consistency across independent metrics — growth, clicks, and content quality — is a stronger signal than any one metric alone, and suggests Instagram's performance is broad-based rather than driven by a single outlier post or metric.

Pattern 2 — YouTube's Engagement Lead Is Efficiency, Not Scale

YouTube has the lowest average daily reach of the four platforms (30,436) yet the highest engagement rate (8.97%). Combined with its mid-pack click share (23.9%, Track D), this suggests YouTube's content resonates strongly with the audience it does reach — the open question is whether that engagement rate would hold if reach were increased, which the current data cannot answer on its own.

Pattern 3 — Volume and Sentiment Are Telling Two Different Stories

Every volume-based metric in this report — clicks, likes, follower growth — points to a healthy, active audience. But sentiment sits at almost exactly 50/50. High engagement volume with neutral-to-negative sentiment is a pattern worth taking seriously: it means the content is being seen and acted on, but not necessarily well-received, and the two should not be conflated when reporting "performance" to stakeholders.

Risk to Note

Because the platform-content matrix in the dashboard export is truncated to 11 of 16 combinations (Section 2), the claim that "Instagram Article is the best-performing content type" is scoped to the visible data. If one of the five hidden combinations — particularly Instagram-Story, given Instagram's overall strength — outperforms it on likes or clicks, that ranking would change. This should be re-verified against the full, unfiltered table before being used in a stakeholder-facing recommendation.

Strategic Recommendations

1 Prioritise Instagram Article-Style Content in the Next Content Cycle

Instagram Article leads all visible content types on likes, comments, clicks, and shares simultaneously. Before scaling this format further, pull the full (untruncated) platform-content table to confirm this ranking holds against the five combinations not visible in this export — particularly Instagram-Story, the one missing Instagram row.

2 Test Increased Reach on YouTube to Validate the Engagement-Rate Advantage

YouTube's 8.97% engagement rate is the highest of the four platforms, but it is currently achieved on the lowest average daily reach (30,436). A controlled reach increase — paid promotion or posting-frequency test — would clarify whether this efficiency holds at scale or was partly a function of a smaller, more self-selected audience.

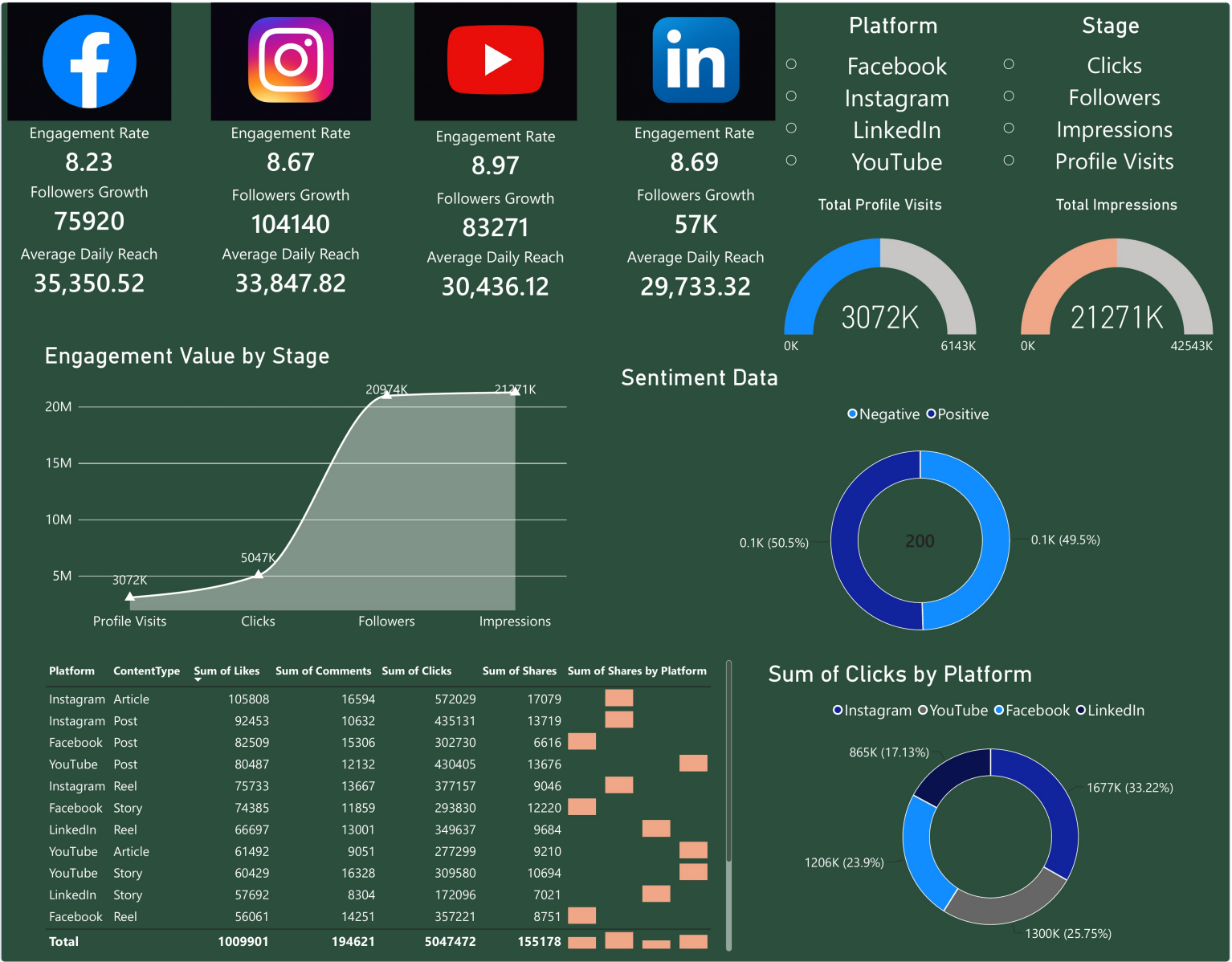
3 Investigate the Drivers Behind the Near-Even Sentiment Split

With sentiment at 50.5% negative against a backdrop of strong engagement volume across every platform, the next analytical step is to break sentiment down by platform and content type (the current dashboard does not do this) to identify whether negative sentiment is concentrated in a specific format or platform, or spread evenly — the two scenarios call for very different responses.

Visual Appendix

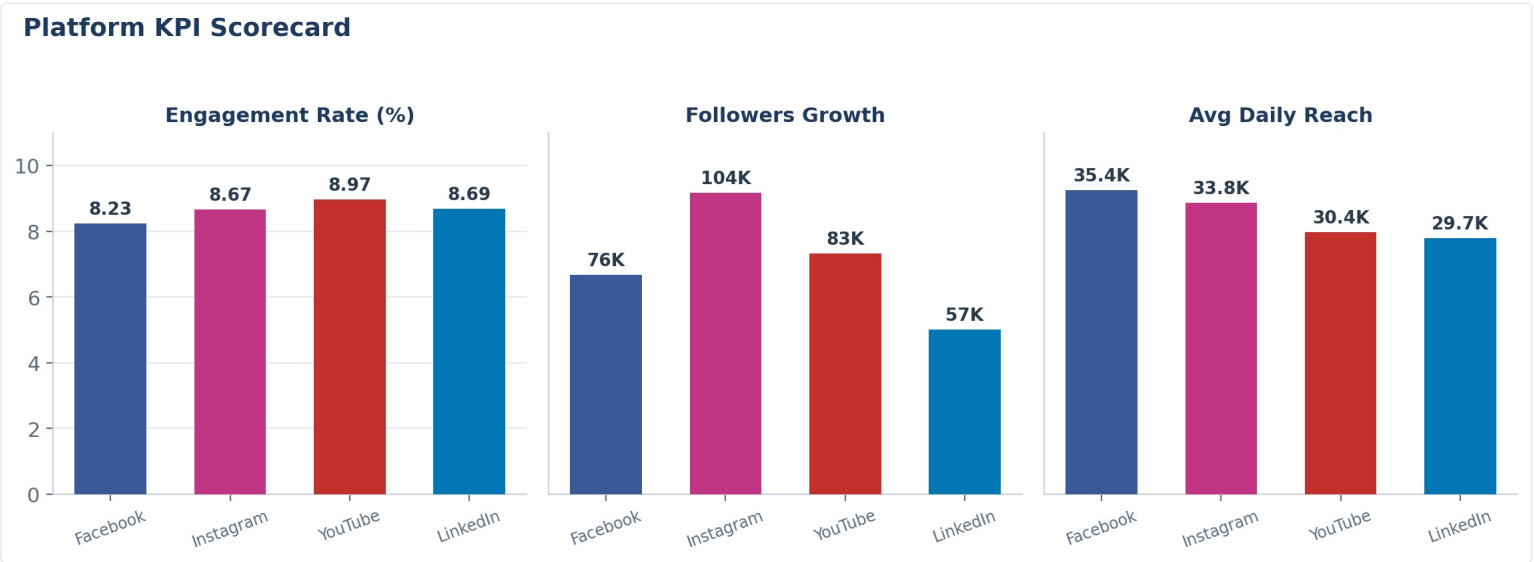
All supporting charts referenced in Section 4, plus the original source dashboard, are reproduced below at full size for reference.

Figure A1 — Original Source Dashboard (Power BI)



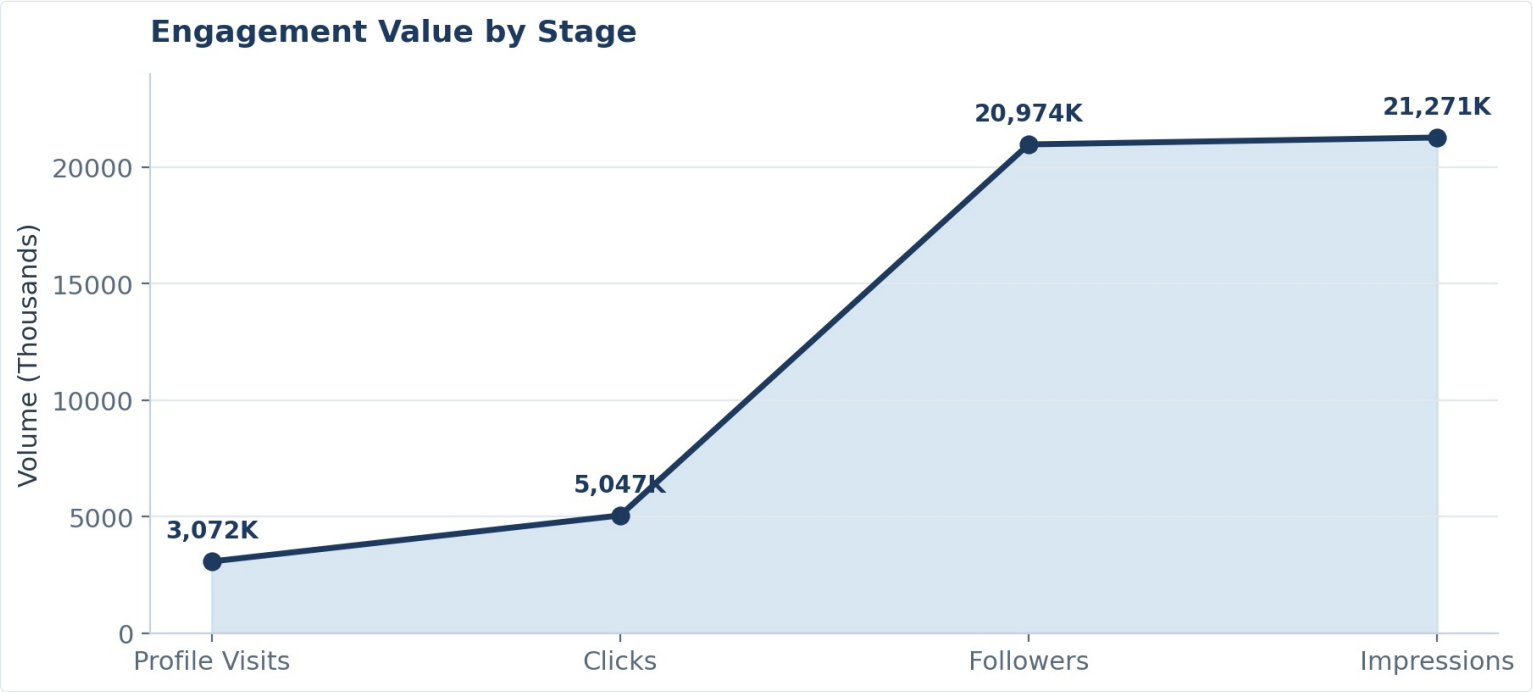
Power BI dashboard as built for this project, showing all platform KPI cards, the engagement-by-stage chart, sentiment donut, content-type matrix, and click distribution by platform.

Figure A2 — Platform KPI Scorecard



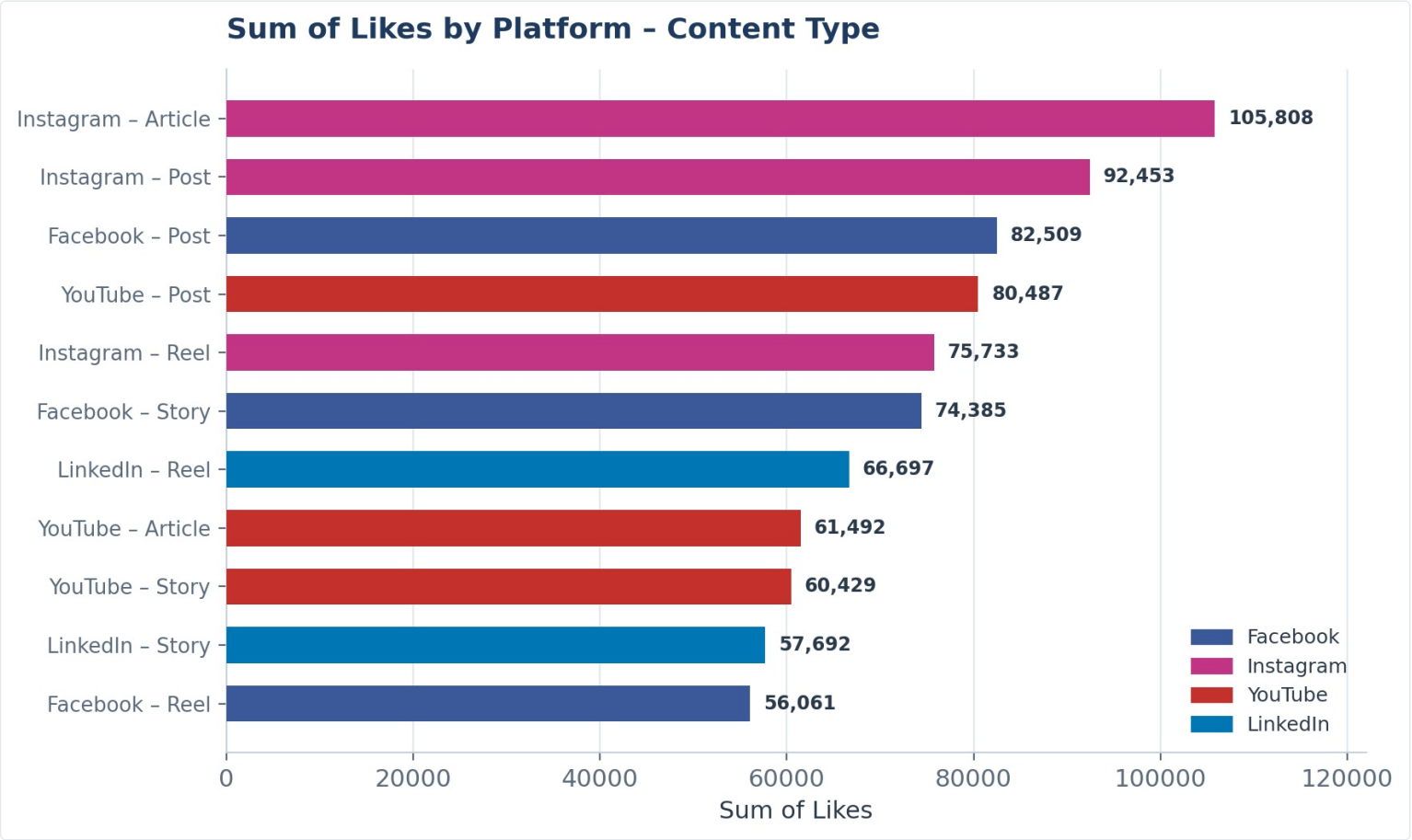
Referenced in Section 4, Track A.

Figure A3 — Engagement Value by Stage



Referenced in Section 4, Track B.

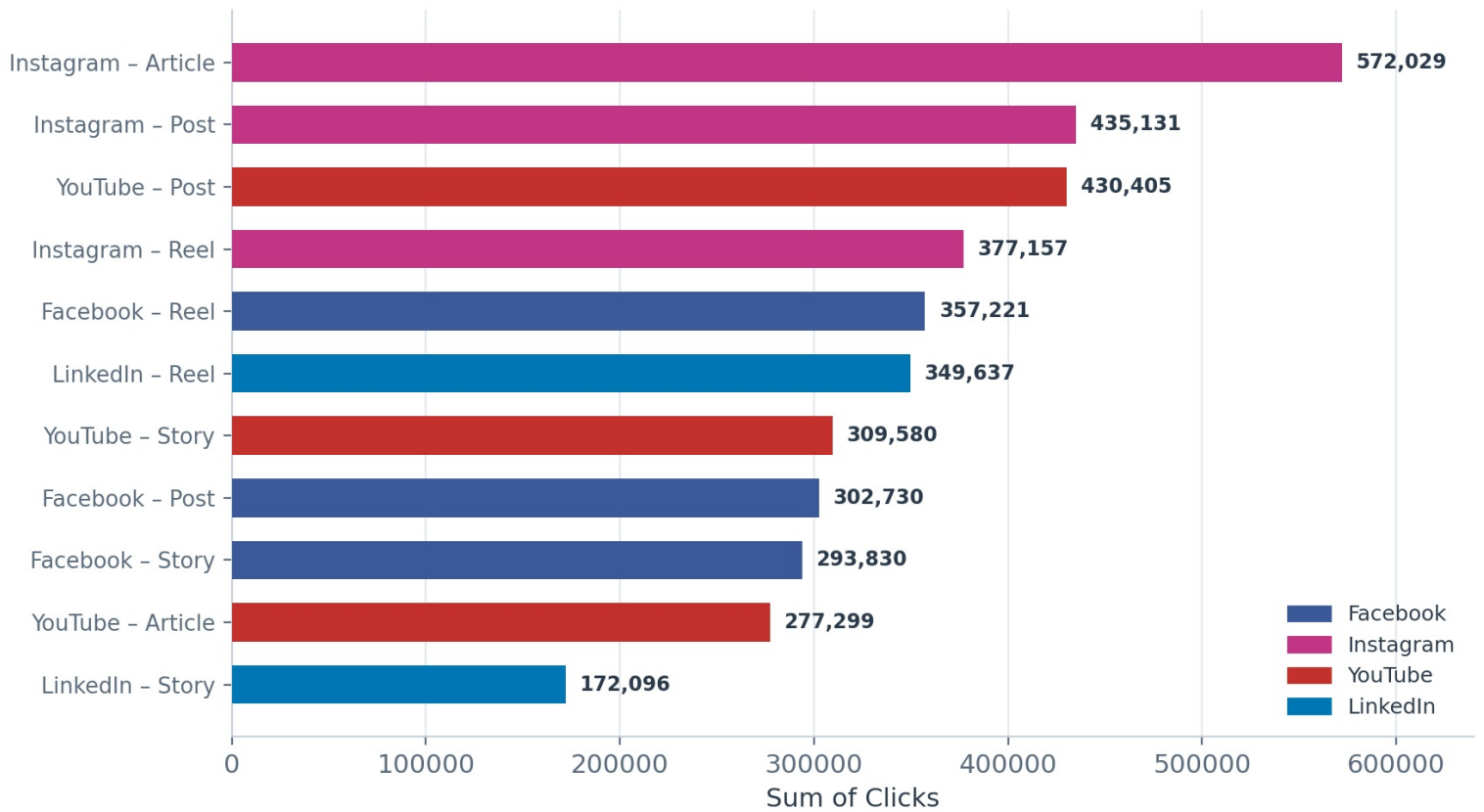
Figure A4 — Sum of Likes by Platform - Content Type



Referenced in Section 4, Track C.

Figure A5 — Sum of Clicks by Platform - Content Type

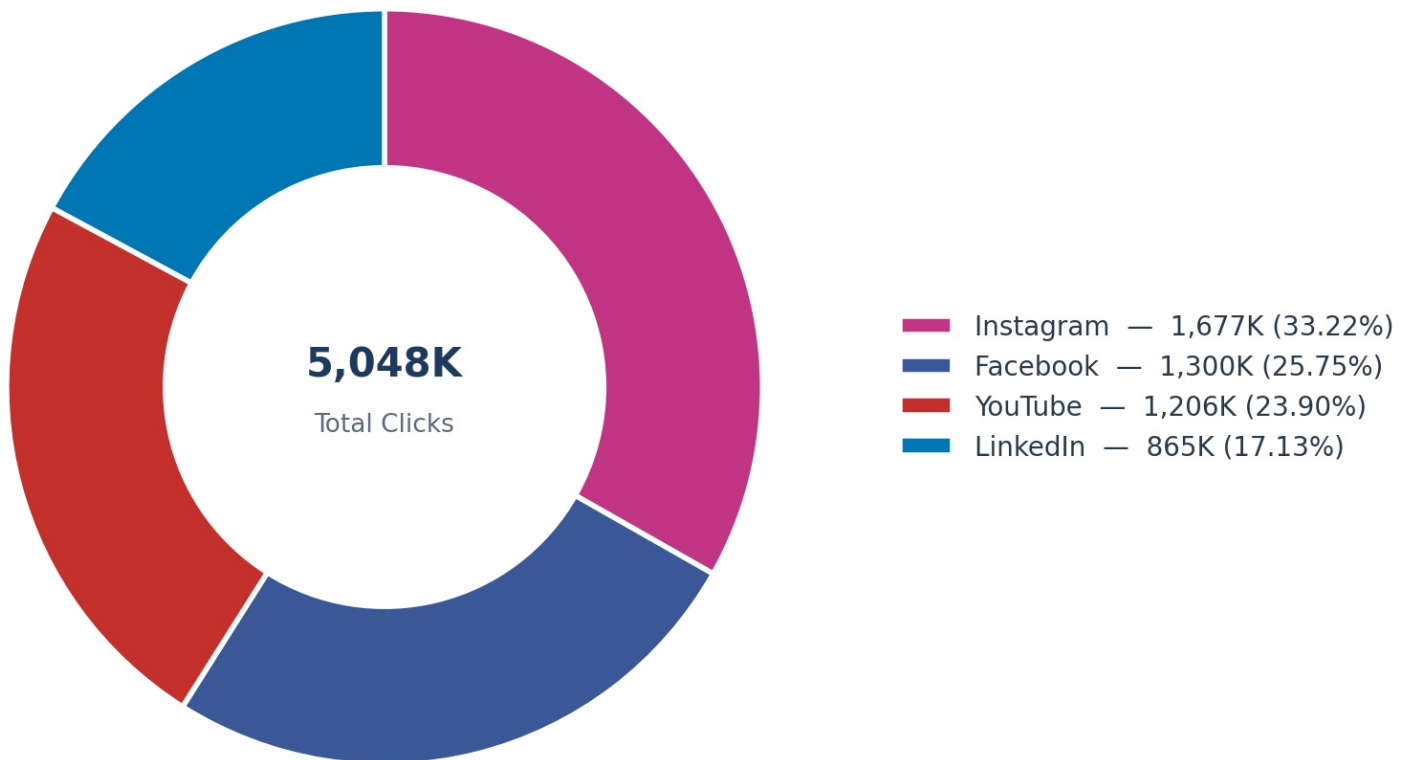
Sum of Clicks by Platform - Content Type



Referenced in Section 4, Track C.

Figure A6 — Sum of Clicks by Platform

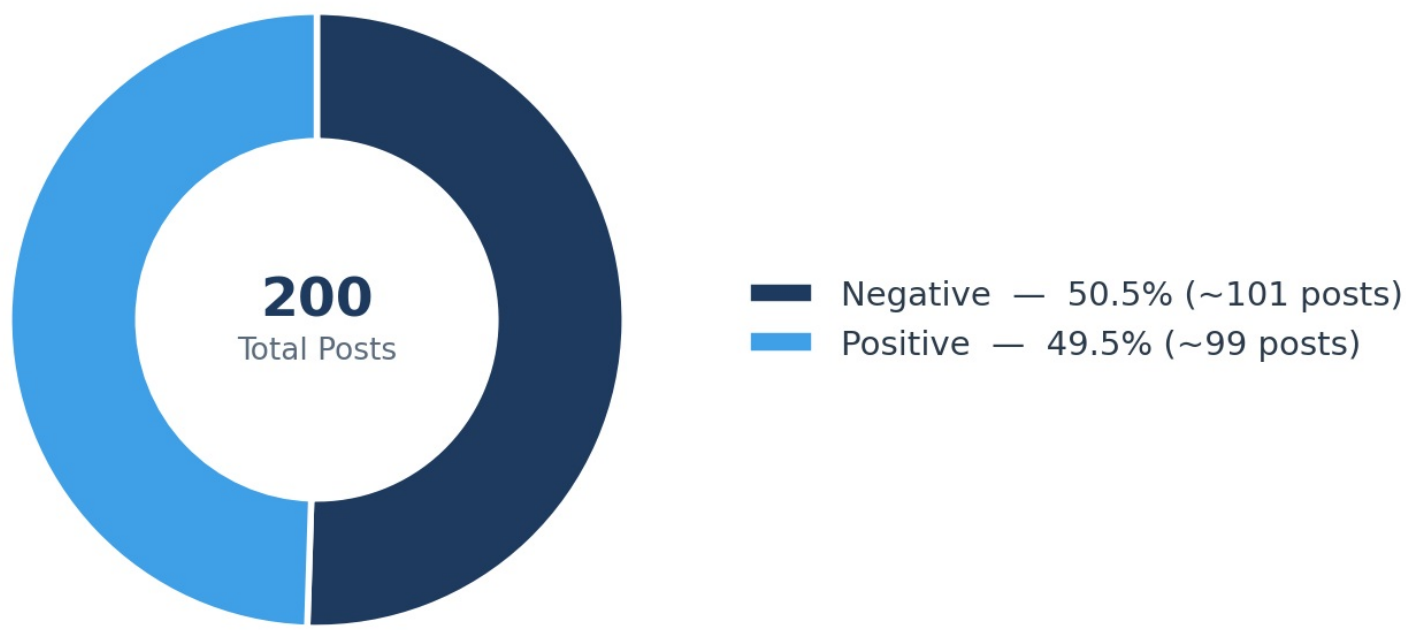
Sum of Clicks by Platform



Referenced in Section 4, Track D.

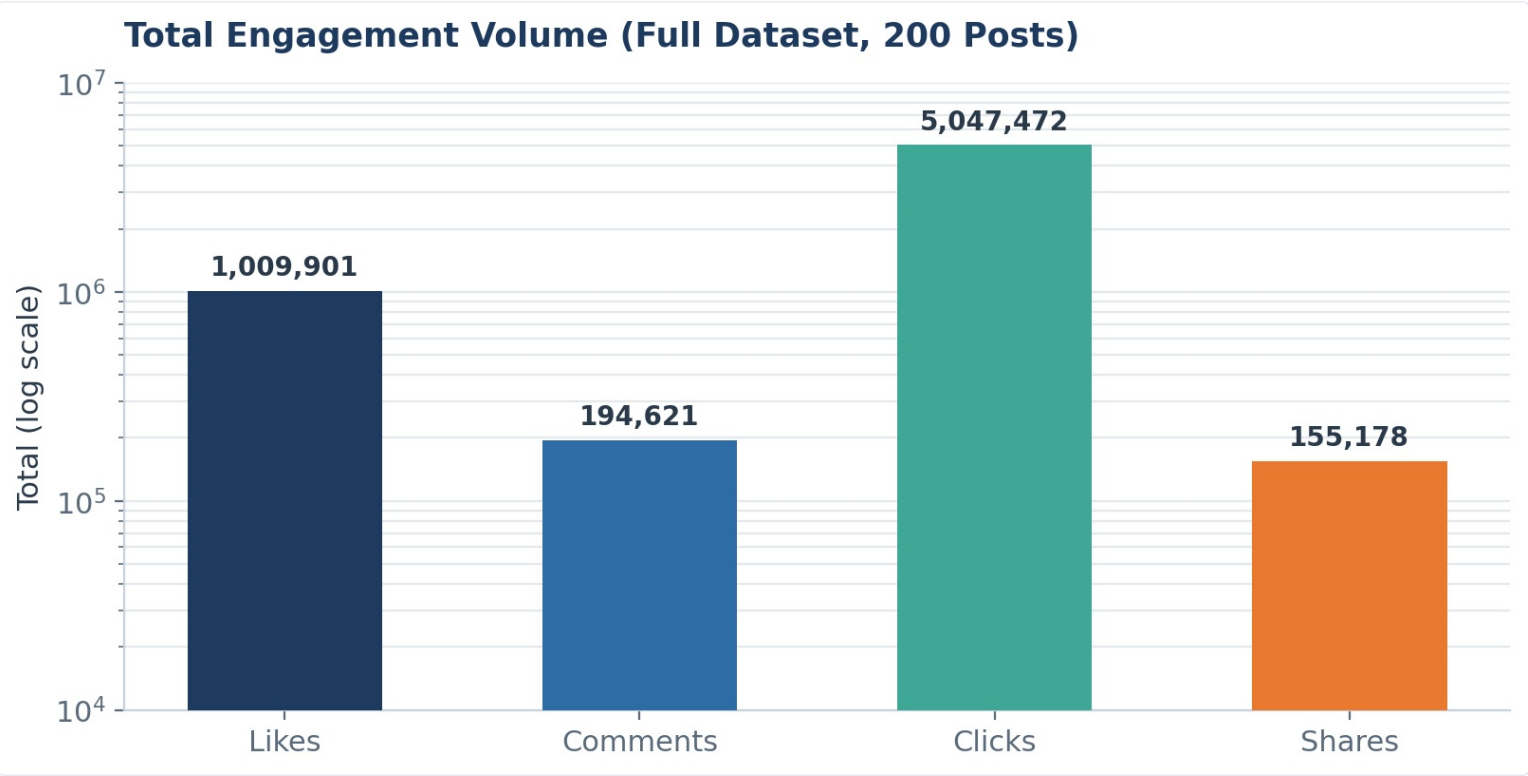
Figure A7 — Sentiment Distribution

Sentiment Distribution



Referenced in Section 4, Track E.

Figure A8 — Total Engagement Volume (Log Scale)



Dataset-wide totals for Likes, Comments, Clicks, and Shares, referenced in Section 2's consistency check.

Methodological Note

This project followed a five-stage SQL-to-dashboard workflow outlined in Section 3: descriptive aggregation, comparative rate calculation, conditional filtering, in-group ranking, and visualisation. All 25 SQL queries were written against the single-table schema described in Section 2; the full query script accompanies this report as a companion file. Every figure presented here was cross-checked against the Power BI dashboard's own totals — for example, the Clicks total derived from the content-type matrix (5,047,472) was checked against the Engagement Value by Stage chart's Clicks figure (5,047K) and the Sum of Clicks by Platform donut ($\approx$ 5,048K), all of which agree to the nearest thousand.

Limitations. The platform-content matrix visual in the dashboard export is truncated to 11 of 16 combinations (Section 2), so Track C's content-type findings are scoped to the visible data and flagged accordingly wherever used. The dashboard does not break sentiment down by platform or content type, so Pattern 3 in Section 5 and Recommendation 3 in Section 6 identify this as a gap for future analysis rather than a finding this report can make directly. Finally, "Followers Growth" is reported exactly as labelled on the source dashboard without independently confirming whether it represents a sum or an average, since the dashboard itself does not specify.

This approach supports structured reasoning and reproducibility: every number in this report can be traced back to either a value shown on the dashboard or a specific query in the accompanying SQL script, so the analysis can be re-run and independently verified against the full dataset.